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CH408S – INTRODUCTION TO MOLECULAR STRUCTURE

Time : 3 hrs **Max Marks :75**

SECTION – A (15X1=15)

Answer **ALL** Questions

I. Choose the correct answer

1. Ψ^2 represents _____.
   a) Probability density b) Amplitude
   c) Linear operator d) Hermitian operator

2. Which of the following atomic orbital is not possible?
   a) 3d b) 4p c) 5g d) 4d

3. The hybridization in SF_6 molecule is
   a) sp^3 b) sp^3d^2 c) sp^3d d) sp^3d^3

4. The bond order of oxygen molecule is
   a) 2.5 b) 1 c) 3 d) 2

5. The modern version of the Gouy balance (SQUID) is used for the determination of _____.
   a) Dipole moment b) Magnetic susceptibility
   c) Refractive index d) Dielectric constant

6. The specific magnetic susceptibility is defined as _____.
   a) $\chi_{sp} = \chi_m/M$ b) $\chi_m = \chi_m/M$
   c) $\chi_{sp} = V_m\chi_m/M$ d) $\chi_{sp} = V_m\chi$

7. σh represent _____ plane of symmetry.
   a) Vertical b) Horizontal c) Diagonal d) Spherical

8. Graphite is an example of _____ crystal.
   a) Covalent b) Molecular c) Ionic d) Metallic

9. The criterion for a molecule to be IR active is
   a) Permanent dipole moment b) Change in dipole moment
   c) Symmetrical d) None

10. Rotational spectra are observed in
    a) IR region b) MW region
    c) UV region d) Radio frequency

II. Match the following

11. Dual character - $\mu = 0$

12. Ammonia - C_2 group

13. 1,2-dichloroethane (cis) - sp^3

14. Identity - de-Broglie

15. CH_4 - Symmetry element

- $\mu = 1.86$ D

- Microwave-inactive

SECTION – B (10X2=20)

Answer any **TEN** Questions

16. Calculate the wavelength associated with an electron (mass 9.1×10^{-31} kg) moving with a velocity of 10^3 m sec^{-1} ($h = 6.626 \times 10^{-34}$ kg m^2 sec^{-1}).

17. Write born interpretation of wave function?

18. What is meant by sp^3 hybridization? Give two examples.

19. Define bond order.

20. Explain the terms: orientation polarizability and distortion polarizability.

21. What information regarding the structure of molecules can be obtained from dipole moments?

22. What is called improper axis of rotation?

23. Identity the point group for the following molecules and list the symmetry elements.
    a) H_2O_2 b) *Trans* CHCl=CHCl

24. Define the following: a) unit cell b) Space lattice

25. Which of the following molecules will show a pure rotational (microwave) spectrum and why: CO, CH_3Cl, H_2O (liquid), NH_3, HCl and NH_4Cl (solid)?

26. State Frank Condon principle.

27. Calculate the energy gap between the rotational levels J=5 and J=6.

SECTION- C (5X8=40)

Answer any **FIVE** Questions

28. Draw and explain the wave functions for s and p - orbitals.

29. Illustrate sp^2 and sp^3 hybridization with suitable example. **(4+4)**

30. Draw the MO diagram for nitrogen molecule and calculate the bond order.

31. Explain how the dipole moment of a gaseous molecule can be determined by the Debye equation?

32. a) What are the postulates of a group? **(3+5)**
    b) Discuss the plane of symmetry and proper axis of rotation with two examples each.

33. a) Differentiate between amorphous and crystalline solids. **(4+4)**
    b) Calculate the number of atoms present in a) FCC & b) BCC.

34. Discuss the rotational spectrum of a rigid diatomic molecule and also the effect of isotopic substitution. **(6+2)**

35. a) Explain the selection rule for the IR and Raman spectra. **(2+6)**
    b) How can we determine the vibrations of polyatomic molecules?

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